

Proof of Phase II

Thm: f is m -strongly convex, $\Leftrightarrow \nabla^2 f(x) \succeq \frac{m}{2} I$
+ twice diff'ble
 $(\nabla^2 f(x) - \frac{m}{2} I) \preceq 0$

Hessian is L -Lipschitz:

$$\|\nabla^2 f(x) - \nabla^2 f(y)\| \leq L \cdot \|x - y\|_2$$

$$\text{and: } \|x_0 - x^*\|_2 \leq \frac{m}{2L}$$

$$\text{pf: FTC: } \int_0^1 \left(\frac{d}{dt} F(t) \right) dt = F(1) - F(0)$$

This tells us:

$$\begin{aligned} \nabla f(x+d) - \nabla f(x) &= \int_0^1 \frac{d}{dt} (\nabla f(x+td)) dt \\ &= \int_0^1 \nabla^2 f(x+td) d dt \end{aligned}$$

We will apply this: $x \leftarrow x^*$, $d \leftarrow x_k - x^*$

Proof of Phase II - continued -

Using expression on prev slide w/ $x = x^*$, $d = x_k - x^*$

$$\Rightarrow \nabla f(x_k) - \cancel{\nabla f(x^*)} = \int_0^1 \nabla^2 f(x^* + t(x_k - x^*)) (x_k - x^*) dt$$

$$x_{k+1} = x_k - (\nabla^2 f(x_k))^{-1} \nabla f(x_k)$$

$$\begin{aligned} \Rightarrow x_{k+1} - x^* &= (x_k - x^*) - (\nabla^2 f(x_k))^{-1} \left[\int_0^1 \nabla^2 f(x^* + t(x_k - x^*)) (x_k - x^*) dt \right] \\ &\quad \hookrightarrow (\nabla^2 f(x_k))^{-1} \nabla^2 f(x_k) \cdot \int_0^1 (x_k - x^*) dt \\ &= (\nabla^2 f(x_k))^{-1} \int_0^1 \left[\nabla^2 f(x_k) - \nabla^2 f(x^* + t(x_k - x^*)) \right] (x_k - x^*) dt \end{aligned}$$

$$\text{Recall: } \|AB\| \leq \|A\| \cdot \|B\| ; \quad \left\| \int (\quad) \right\| \leq \int \| \quad \|$$

$$\begin{aligned} \|x_{k+1} - x^*\| &\leq \|\nabla^2 f(x_k)^{-1}\| \cdot \int_0^1 \underbrace{\|\nabla^2 f(x_k) - \nabla^2 f(x^* + t(x_k - x^*))\|}_{\substack{\boxed{(1-t) \cdot \|x_k - x^*\|} \\ \nabla^2 f \text{ is } L\text{-Lipschitz}}} dt \cdot \|x_k - x^*\| \\ &\leq \|\nabla^2 f(x_k)^{-1}\| \cdot \int_0^1 L \cdot \boxed{(1-t) \cdot \|x_k - x^*\|} \cdot dt \cdot \|x_k - x^*\| \end{aligned}$$

Proof of Phase II - continued -

So far, we have:

$$\|x_{k+1} - x^*\| \leq \underbrace{\|\nabla^2 f(x_k)^{-1}\|}_{\downarrow} \cdot \underbrace{\int_0^1 L(1-t) dt}_{L \int_0^1 (1-t) dt = L \cdot \frac{1}{2}} \cdot \|x_k - x^*\|^2$$

If $A \neq 0$ then $\|A\| = \lambda_{\max}(A)$

$A \neq 0$ then $\|A^{-1}\| = \lambda_{\max}(A^{-1}) = \frac{1}{\lambda_{\min}(A)}$

f m-strongly convex $\Rightarrow \lambda_{\min} \nabla^2 f(x_k) \geq \frac{m}{2}$

$\Rightarrow$ as $\nabla^2 f(x_k)$

Putting it all together:

$$\begin{aligned} \|x_{k+1} - x^*\| &\leq \frac{2}{m} \cdot L \cdot \frac{1}{2} \cdot \|x_k - x^*\|^2 \\ &= \frac{L}{m} \|x_k - x^*\|^2 \quad \text{as claimed.} \end{aligned}$$

Newton's Method: Summary

More assumptions: also require f is twice differentiable
and $\nabla^2 f$ is Lipschitz

⑧ More computation: need to compute inverse of $(\nabla^2 f(x))$

very fast - quadratic-convergence.

$$O(\log \log (1/\epsilon))$$

$$\text{i.e., error} \sim \left(\frac{1}{2}\right)^{2^t}$$